Chronic Kidney Disease (CKD)

Karen Pendergrass, Giorgos Aristotelous

Microbiome Signatures, Microbiome Medicine

Abstract. Chronic kidney disease (CKD) microbiome signature, key dysbiosis patterns, uremic toxins, and targeted interventions supporting renal health.

Published 2025-11-25 · https://microbiomemedicine.com/conditions/chronic-kidney-disease-ckd · A Microbiome Signatures condition monograph

Overview

Dysbiosis in chronic kidney disease (CKD) reflects a shift toward reduced beneficial taxa and increased pathogenic, uremic toxin-producing species, driven by a bidirectional interaction in which the uremic environment disrupts microbial composition and dysbiotic metabolites accelerate renal deterioration. [1][2] Loss of microbial diversity, diminished short-chain fatty acid (SCFA)-producing genera such as Faecalibacterium, Roseburia, and Ruminococcus, and increased indoxyl sulfate and p-cresyl sulfate producers heighten intestinal permeability, systemic inflammation, and endotoxemia, all of which intensify kidney injury. [3] Evidence from fecal microbiota transplantation studies demonstrates that CKD-derived microbiota can directly promote renal fibrosis, supporting a dose-response relationship where the abundance of uremic toxin precursor species rises with advancing CKD and drives progressive renal damage. [4]

Associated Conditions

Associated conditions in chronic kidney disease encompass a broad spectrum of interrelated systemic complications that emerge early in renal impairment and intensify as kidney function declines. These complications reflect the kidney's central role in metabolic, endocrine, cardiovascular, neuromuscular, and immunologic homeostasis, and their cumulative burden profoundly shapes patient outcomes. As dysregulated uremic metabolism, chronic inflammation, mineral imbalances, and microbiome disturbances converge, patients develop a characteristic pattern of comorbidities that includes cardiovascular disease, metabolic dysfunction, anemia, mineral bone disorder, neurocognitive decline, sarcopenia, and heightened infection susceptibility. The prevalence and severity of these conditions rise sharply across CKD stages, underscoring the need for early recognition, mechanistic understanding, and integrated therapeutic strategies.

Causes

Chronic kidney disease is increasingly understood as a condition shaped not only by impaired renal filtration but by a complex interplay between dysbiotic microbial communities, disrupted metabolite production, and altered metallomic homeostasis. As renal function declines, shifts in gut microbial

composition generate pathogenic metabolites, weaken mucosal barrier integrity, amplify systemic inflammation, and modify host pathways involved in cardiovascular, metabolic, and neuroimmune regulation. Complementary but distinct conceptual hypotheses have emerged, including metabolic endotoxemia, dysbiosis-driven uremic toxin generation, toxic metal-driven metallomic dysbiosis, essential metal deficiency, TMAO-mediated cardiovascular injury, RAAS dysregulation, and disruption of the tryptophan-AhR axis. Together, these frameworks describe how altered microbial communities, their metabolites, and their metal handling properties converge on shared pathophysiological endpoints, including intestinal barrier failure, chronic inflammation, cardio-renal remodeling, neuropsychiatric impairment, and accelerated CKD progression.

Primer

The interplay between metal homeostasis, microbial dysbiosis, and metabolomic disruptions in chronic kidney disease reveals how systemic and localized factors converge to drive CKD progression. These interrelationships underscore the metallome, metabolome, and microbiome's central role in the disease's pathogenesis and highlight opportunities for microbiome-based diagnostics and microbiome-targeted interventions (MBTIs).

Gut-Kidney Axis

The development of CKD is fundamentally linked to dysbiosis through multiple interconnected mechanisms collectively termed the gut-kidney axis. Significant alterations in gut microbiome composition, richness, diversity, and blood and fecal metabolic composition have been consistently documented in patients with CKD and kidney failure, strongly supporting a crucial role of gut dysbiosis in the pathogenesis of CKD.[5] Studies have demonstrated that gut bacterial dysbiosis contributes to CKD via several critical mechanisms, including accumulation of uremic toxins, decreased production of short-chain fatty acids (SCFAs), disturbed enteroendocrine signaling, and development of a leaky gut barrier.[6]

Metabolomic Signature of CKD

The metabolomic signature of chronic kidney disease represents a complex constellation of altered metabolites reflecting impaired kidney clearance, intestinal dysbiosis, and systemic metabolic derangement. This signature includes elevated protein-derived uremic toxins (PCS, IS, PS, TMAO), small molecular weight retention solutes (urea, creatinine), accumulation of heavy metals, amino acid imbalances, bile acid alterations, and deficiency in beneficial short-chain fatty acids (SCFAs). The interplay between these metabolomic changes drives disease progression through multiple pathways involving inflammation, oxidative stress, fibrosis, and vascular dysfunction.[7] Understanding and modulating this metabolomic signature represents a promising therapeutic frontier for slowing CKD progression and improving patient outcomes.

Mismetallation

Mismetallation represents a central yet underrecognized mechanism accelerating the progression of chronic kidney disease by allowing toxic heavy metals to displace essential metal cofactors within metalloproteins, metalloenzymes, and mitochondrial complexes. In CKD, impaired renal clearance amplifies the accumulation and bioactivity of cadmium, lead, mercury, arsenic, and chromium, all of which

use ionic mimicry to substitute for physiologic metals such as zinc, iron, copper, manganese, and calcium. This displacement collapses antioxidant enzyme systems, impairs mitochondrial ATP generation, and initiates cell death pathways including apoptosis and ferroptosis. The renal proximal tubule, with its high metabolic demand and reliance on metal-dependent enzymatic and mitochondrial functions, is uniquely vulnerable. Mismetallation therefore acts as a unifying pathological process linking heavy metal exposure to oxidative stress, mitochondrial dysfunction, vascular injury, hypoxia–signaling dysregulation, and progressive nephron loss in CKD.

Nutritional Immunity

Nutritional immunity represents a critical arm of host defense in CKD, wherein essential transition metals, including iron, zinc, and manganese, are actively sequestered to prevent microbial proliferation. This metal-withholding strategy is mediated through a range of high-affinity host proteins, including calprotectin, transferrin, lipocalin-2, and hepcidin, which tightly regulate metal availability in both extracellular and intracellular compartments. However, in chronic kidney disease (CKD), over time, this finely tuned network results in a pathological state characterized by functional iron deficiency, zinc and manganese depletion, which collectively exacerbate anemia, promote tissue damage, and increase susceptibility to infection. Understanding how CKD results in these elevated nutritional immunity factors offers key insights into the disease's complex interplay between immune dysfunction, mineral metabolism, host-pathogen interactions, and potential interventions.

Interventions

A growing portfolio of mechanistically diverse interventions now targets the dysbiosis–uremic toxin–metallomic triad that characterizes CKD progression. The following section surveys key candidates that act along the gut–kidney axis and at the level of renal cellular biology, including carbon-based adsorbents that sequester uremic solutes in the intestinal lumen, mitochondrial protective agents that constrain oxidative injury, and prebiotic fibers that redirect nitrogen disposal toward the fecal compartment. Additional strategies encompass immunomodulatory and toxin-lowering proteins, biotherapeutic yeasts, and live bacterial therapeutics that restore barrier integrity, suppress pathobionts, and reduce production and systemic absorption of indoxyl sulfate, p-cresyl sulfate, and related metabolites. Finally, metallomic interventions such as dimethylglyoxime (DMG) nickel chelation directly address heavy metal–driven dysbiosis and enzymatic virulence. Together, these approaches exemplify how targeting microbial metabolism, uremic toxin handling, and metal homeostasis may complement or, in selected contexts, delay conventional renal replacement therapies while mitigating systemic complications of CKD.

References

1. Assembled human microbiome and metabolome in chronic kidney disease: Dysbiosis a double-edged sword interlinking Circ-YAP1, Circ-APOE & Circ-SLC8A1. Z. Ea, A. Sm, A. Nm, and K. Mo. (Toxicol Rep. 2025 May 29;14:102058.)
2. Dietary Intake and Gut Microbiome in Chronic Kidney Disease.. J. Kemp, M. Ribeiro, N. Borges, L. Cardozo, D. Fouque, and D. Mafra,. (American Society of Nephrology. Clinical Journal, Mar. 2025)
3. Gut Microbiome in Patients with Chronic Kidney Disease Stages 4 and 5: A Systematic Literature Review. I. L. Suliman et al.. (International Journal of Molecular Sciences, Nov. 2025)

4. Toxic microbiome and progression of chronic kidney disease: insights from a longitudinal CKD-Microbiome Study. M. Laiola et al.. (Gut, Jun. 2025)
5. Therapeutic Potential of Photobiomodulation for Chronic Kidney Disease.. Ji Bian, Ann Liebert, et al.. (International Journal of Molecular Sciences. 2022.)
6. Therapeutic Potential of Photobiomodulation for Chronic Kidney Disease.. Ji Bian, Ann Liebert, et al.. (International Journal of Molecular Sciences. 2022.)
7. Therapeutic Potential of Photobiomodulation for Chronic Kidney Disease.. Ji Bian, Ann Liebert et al.. (International Journal of Molecular Sciences. 2022.)

Generated from microbiomemedicine.com/conditions/chronic-kidney-disease-ckd. © Microbiome Signatures.